

ADI VAMSI SAI MADDIRALA

AI Engineer | Backend Engineer | Python, APIs, LLM Workflows, Automation | Building Production-Grade AI Systems

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TARGET ROLES

AI Engineer | Applied AI Engineer | GenAI Engineer | LLM Application Engineer | Backend Engineer | Software Engineer

SUMMARY

Python and backend developer focused on building APIs, automation workflows, and AI-enabled systems that solve real business problems. Experience spans Python backend development, LLM API integrations, workflow automation, REST APIs, and production reliability improvements. Independent AI projects include Indian SME Engine, Agent Hub, and applied AI apps focused on RAG, Text-to-SQL, prompt chaining, and agentic workflows.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Backend & Data: REST APIs, Microservices, FastAPI, Flask, Spring Boot, Node.js, Express, PostgreSQL, Prisma ORM, Hibernate/JPA, WebSockets

AI / LLM Engineering: OpenAI API, Claude, Gemini, LangChain, LangGraph, RAG, Prompt Engineering, AI Agents, Agentic Orchestration, embeddings, vector search, ChromaDB, Ollama, PyTorch

Developer Tools: Cursor, Google Antigravity, Claude Code, Codex, Git, Docker, Linux, AWS, Streamlit, Agile/Scrum

EXPERIENCE

Python Developer | Capgemini | Contract

Jul 2024 - May 2026 | San Antonio, TX / Remote

- Built and maintained Python backend services for automation, data processing, API integrations, and internal business workflows used by cross-functional teams
- Improved backend processing speed by 20% by optimizing SQL queries, refactoring service logic, and reducing unnecessary API calls.
- Integrated LLM APIs into classification and extraction workflows, reducing repetitive manual review effort by 35%.
- Added structured logging, exception handling, input validation, and retry logic across backend modules, lowering recurring production issues by 15%.
- Reduced critical backend issue resolution time from 48 hours to 12 hours through debugging workflows, sprint delivery, and release support.
- Collaborated with product, QA, and engineering teams to deliver backend fixes, API enhancements, and automation features in Agile sprint cycles.

Software Engineer | New Relic

Jun 2020 - Dec 2022 | Hyderabad, India

- Built Java 8/11 Spring Boot backend services for an observability platform supporting high-volume queries across distributed microservices.
- Designed and optimized REST API contracts, reducing inter-service latency by approximately 15% through payload restructuring.
- Tuned Hibernate ORM queries and database interaction patterns to improve backend throughput and reliability under sustained production load.
- Automated data validation, log analysis, and API regression testing with Python scripts, reducing manual QA cycle time by approximately 30%.
- Refactored legacy backend components and expanded critical-path test coverage, reducing page load times by approximately 10%.

PROJECTS

Indian SME Engine | Node.js, Express, PostgreSQL, Prisma, WebSockets, JWT, LLM APIs

GitHub: <https://github.com/AdiVamsi/indian-sme-engine>

Built an LLM-powered multi-tenant CRM for Indian SMBs that captures leads from web forms and WhatsApp, classifies and scores them with AI, and surfaces prioritized follow-ups through a real-time operator dashboard.

- Engineered JWT auth, tenant data isolation, WebSocket updates, lead APIs, AI scoring logic, and an admin action queue.
- Designed the system around practical business workflows: lead intake, qualification, prioritization, follow-up automation, and operator handoff.

Agent Hub | Python, Claude, Codex, AutoResearch, Agentic Orchestration

GitHub: <https://github.com/AdiVamsi/agent-hub>

Built a self-improving agent platform where agents optimize tasks through programmatic artifact generation, scalar scoring, and git commit/reset loops.

- Designed a meta-agent workflow for overnight optimization loops with clean git history.
- Made it compatible with Claude, Codex, OpenAI, Gemini, and other LLM providers.

AI Apps Portfolio | Streamlit, RAG, ChromaDB, Text-to-SQL, OpenAI/Ollama

Built a collection of applied AI apps including a chatbot, prompt chaining app, YouTube summarizer, guarded Text-to-SQL app, and multi-document RAG system.

- Designed reusable LLM client patterns for provider switching.
- Added transcript processing, SQL guardrails, retrieved-chunk transparency, and document citation behavior.

EDUCATION

Webster University | San Antonio, TX

Master's degree, Information Technology

Jan 2023 - May 2024 | Grade: 3.6/4.0

GITAM Deemed University | Hyderabad, India

Bachelor's degree, Computer Science Engineering

Jun 2016 - Jun 2020